

Diary entry for Spectral-Sequences

Aditya Pahuja

2026-06-29

Contents

1	Theory	2
2	Examples and applications	2
2.1	Some results in homological algebra	2
2.2	Serre spectral sequence	3
3	Practice	5

1 Theory

2 Examples and applications

2.1 Some results in homological algebra

Proposition 2.1 (Four lemmas). Consider the following commutative diagram of modules (or just in some abelian category), in which the two rows are exact:

$$\begin{array}{ccccccc} A & \longleftarrow & B & \longleftarrow & C & \longleftarrow & D \\ \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \downarrow \delta \\ A' & \longleftarrow & B' & \longleftarrow & C' & \longleftarrow & D' \end{array}$$

- (a) If α and γ are monomorphisms and δ is an epimorphism, then β is an monomorphism.
- (b) If β and δ are epimorphisms and α is an monomorphism, then γ is a epimorphism.

(The diagram is drawn backwards to emphasize how the double complex is oriented in the plane.)

Proof. By extending via zeroes, we can view the given diagram as a first quadrant double complex E which is concentrated in finitely many points of the grid, so the corresponding (starting-at-horizontal or starting-at-vertical) spectral sequences converge to the homology of its total complex. If we take horizontal homology first to get to E^1 , then the E^1 (hence E^∞) page is all zeroes, so the spectral sequence must converge to zero.

Now we consider the two parts separately and pass to E^1 by taking vertical homology.

- (a) The E^1 page is

$$\begin{array}{ccccccc} 0 & \xleftarrow{d^1} & \ker \beta & \xleftarrow{d^1} & 0 & & \ker \delta \\ & & & & \searrow d^2 & & \\ \text{coker } \alpha & & \text{coker } \beta & & \text{coker } \gamma & & 0 \end{array}$$

From here on out, the differentials into and out of $\ker \beta$ are all zeroes: in E^1 , the differentials are the adjacent zero maps; in E^2 , the incoming differential is from $\text{coker } \delta = 0$ and the outgoing one goes outside the first quadrant; and in the remaining pages the differentials come from or end at a place outside the first quadrant. This means $\ker \beta$ persists in E^∞ , which we have already discovered consists of all zeroes, so $\ker \beta = 0$.

- (b) Similarly, the E^1 page in this case is

$$\begin{array}{ccccccc} 0 & & \ker \beta & & \ker \gamma & & \ker \delta \\ & & \searrow d^2 & & & & \\ \text{coker } \alpha & & 0 & \xleftarrow{d^1} & \text{coker } \gamma & \xleftarrow{d^1} & 0 \end{array}$$

and the differentials involving $\text{coker } \gamma$ from here on out are all zeroes, meaning $\text{coker } \gamma$ persists to E^∞ , hence is zero. $\square$

Exercise 2.2. Prove the five lemma using a similar spectral sequence argument. Recall that the five lemma says that for a commutative diagram

$$\begin{array}{ccccccccc} A & \longleftarrow & B & \longleftarrow & C & \longleftarrow & D & \longleftarrow & E \\ \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \downarrow \delta & & \downarrow \epsilon \\ A' & \longleftarrow & B' & \longleftarrow & C' & \longleftarrow & D' & \longleftarrow & E' \end{array}$$

in which the two rows are exact, β and δ are isomorphisms, δ is an epimorphism, and α is a monomorphism, we have that γ is an isomorphism. (In real life this often just manifests as the outer four maps being isomorphisms implying that γ is an isomorphism.)

Proposition 2.3 (Snake lemma). Consider the commutative diagram of modules

$$\begin{array}{ccccccc} 0 & \longleftarrow & C & \xleftarrow{g} & B & \xleftarrow{f} & A \\ & & \downarrow \gamma & & \downarrow \beta & & \downarrow \alpha \\ & & C' & \xleftarrow{g'} & B' & \xleftarrow{f'} & A' \longleftarrow 0 \end{array}$$

in which the rows are exact. There is an exact sequence

$$\ker f \rightarrow \ker \alpha \rightarrow \ker \beta \rightarrow \ker \gamma \xrightarrow{\partial} \operatorname{coker} \alpha \rightarrow \operatorname{coker} \beta \rightarrow \operatorname{coker} \gamma \rightarrow \operatorname{coker} g'.$$

Proof. As before, complete the diagram with zeroes to make it into a first quadrant double complex E . Taking horizontal homology gives an E^1 page of

$$\begin{array}{ccc} 0 & 0 & \ker f \\ \operatorname{coker} g' & 0 & 0 \end{array}$$

after which the spectral sequence stabilizes, so this is equal to the E^∞ page.

Taking vertical homology instead gives an E^1 page of

$$\begin{array}{ccc} \ker \gamma & \ker \beta & \ker \alpha \\ \operatorname{coker} \gamma & \operatorname{coker} \beta & \operatorname{coker} \alpha \end{array}$$

Passing to E^2 by taking horizontal homology must kill $\ker \beta$ and $\operatorname{coker} \beta$ because in future pages they are not involved in nonzero differentials. For a similar reason, the homology at $\ker \alpha$ must be $\ker f$ and the homology at $\operatorname{coker} \gamma$ must be $\operatorname{coker} g'$, since the objects at those locations also persist from E^2 to E^∞ . This gives the entire exact sequence except for the connecting homomorphism, which we consider now. In the E^2 page the only nontrivial differential is a map

$$d^2: \ker d_{2,0}^1 \rightarrow \operatorname{coker} d_{1,1}^1,$$

which must be an isomorphism because the objects at these locations must vanish in the E^∞ page and they will not be involved in nontrivial differentials in the future. Define the connecting map ∂ as the composition

$$\ker \gamma \twoheadrightarrow \operatorname{coker} d_{1,1}^1 \xrightarrow{\sim} \ker d_{2,0}^1 \hookrightarrow \operatorname{coker} \alpha,$$

inverting d^2 for the middle map. The kernel of ∂ is that of the first map in the composition, namely the image of the map $d_{1,1}^1: \ker \beta \rightarrow \ker \gamma$ in the exact sequence, so $\ker \beta \rightarrow \ker \gamma \xrightarrow{\partial} \operatorname{coker} \alpha$ is exact at $\ker \gamma$. Similarly, the image of ∂ is isomorphic to the kernel of the map in the exact sequence $d_{2,0}^1: \operatorname{coker} \beta \rightarrow \operatorname{coker} \alpha$, proving exactness at $\operatorname{coker} \alpha$. $\square$

Exercise 2.4. Use essentially the same argument to show that a short exact sequence of chain complexes induces a long exact sequence in homology.

Note that in all the examples above, the total complex itself is essentially irrelevant; the only important thing is that both spectral sequences furnished by the double complex converge to the same thing (which fortunately was all zeroes).

2.2 Serre spectral sequence

Theorem 2.5 (Serre spectral sequence). Let $F \hookrightarrow E \rightarrow B$ be a fibration with path connected base space B whose fundamental group acts trivially on $H_*(F)$. Then, there is a homological spectral sequence $E_{p,q}^2 = H_p(B; H_q(F))$ converging to $H_*(E)$ and a cohomological spectral sequence $E_2^{p,q} = H^p(B; H^q(F; R))$ converging to $H^*(E)$ for some abelian group R . Furthermore, if R is a ring, there are bilinear products $E_r^{p,q} \otimes E_r^{s,t} \rightarrow E_r^{p+q, s+t}$ on each page such that

- (a) Each d_r is a derivation, namely $d_r(xy) = (d_r x)y + (-1)^{p+q} x d_r y$ where $x \in E_r^{p+q}$. Note that this means the product on page $r+1$ is induced by the product on page r .
- (b) The product in E^2 is the standard cup product with coefficients in the cohomology ring of the fiber.
- (c) The cup product on $H^*(E)$ induces the products on the E^∞ page.

This is particularly useful when you can say something nice about some of the terms of the spectral sequence, such as knowing that E is contractible or that there are a lot of zero rows/columns (perhaps the spectral sequence is concentrated in just a few rows or columns). Note also that the triviality of the action of π_1 often manifests as X just being simply connected.

Here are a whole bunch of examples.

Theorem 2.6 (Hurewicz's theorem). Suppose that X is n -connected, meaning the first n homotopy groups of X are zero. Then, the first n reduced homology groups (with integer coefficients) of X are also zero, and if $n > 0$, then $\pi_{n+1}(X, x_0) \cong H_{n+1}(X)$. If $n = 0$, we instead have $\pi_1(X, x_0)^{\text{ab}} \cong H_1(X)$, the abelianization of the fundamental group.

Proof. (This proof was shown to me by my friend Nick.)

For $n = 0$ one can prove the theorem manually without too much blood, sweat, or tears; see Hatcher appendix 2A. To extend to larger n , we induct on n . Suppose that X is n -connected for $n \geq 1$ (notably, simply connected), and consider the Serre spectral sequence for the path fibration $\Omega X \rightarrow PX \rightarrow X$. Using the inductive hypothesis and the fact that ΩX is $(n-1)$ -connected, the E^2 page of the Serre spectral sequence looks like

$$\begin{array}{cccccc}
 H_n(\Omega X) = \pi_n(\Omega X) & & & & & \\
 0 & 0 & \cdots & 0 & 0 & \\
 \vdots & \vdots & \ddots & \vdots & \vdots & \\
 0 & 0 & \cdots & 0 & 0 & \\
 \mathbb{Z} & 0 & \cdots & 0 & H_n(X) & H_{n+1}(X)
 \end{array}$$

where rows 1 through $n-1$ are all zeroes because the coefficient group $H_k(\Omega X)$ is zero for $1 \leq k < n$. Since the entire diagonal $p+q = n-2$ consists of zeroes, $H_n(X)$ can only ever map into zero, so it persists in the E^∞ page; since PX is contractible, this means $H_n(X)$ must be zero. To deal with $H_{n+1}(X)$, we note that the entry above $H_n(X)$ is zero by hypothesis if $n > 1$, and if $n = 1$ then this entry sits directly to the right of $\pi_n(X)$, so there is no differential from it into $\pi_n(X)$. Thus the only nontrivial differential involving $H_{n+1}(X)$ or $\pi_n(X)$ is the one in page $n+1$, where $H_{n+1}(X)$ maps into $\pi_n(X)$; this is the only opportunity for these two to become zero (which they must in E^∞), which means that

$$0 \rightarrow H_{n+1}(X) \xrightarrow{d^{n+1}} \pi_n(\Omega X) \rightarrow 0$$

is exact, i.e. $H_{n+1}(X) \cong \pi_n(\Omega X) \cong \pi_{n+1}(X)$. □

Example 2.7. Let $1 \rightarrow A \rightarrow B \rightarrow C \rightarrow 1$ be a short exact sequence of groups. Each homomorphism $B \rightarrow C$ can be realized by a map $K(B, 1) \rightarrow K(C, 1)$. Converting this into a fibration and feeding it into the long exact sequence shows that

$$1 \rightarrow \pi_1(F) \rightarrow B \rightarrow C \rightarrow 1$$

is exact, so $\pi_1(F) = A$.

elaborate

uh ok this is more complicated than i thought

3 Practice

Courtesy of Ajay.

Exercise 3.1. Use the Atiyah-Hirzebruch spectral sequence to compute the K -theory of $\mathbb{C}P^\infty$.

Exercise 3.2. Compute the homology of $K(\mathbb{Z}, 2) = \mathbb{C}P^\infty$ using the Serre spectral sequence, noting that there is a fibration $S^\infty \rightarrow \mathbb{C}P^\infty$ (what is its fiber?). Note that you should be able to verify your answer easily using cellular homology.

Exercise 3.3. Do the same as the above exercise for $\mathbb{R}P^\infty$ (perhaps with $\mathbb{Z}_2$ coefficients; I am not sure how tractable this is).

Exercise 3.4. If $E \rightarrow B$ is a covering space, how are the homology of E and B related?

Exercise 3.5. Deduce the Künneth formula for (co)homology with coefficients in a field using the Serre spectral sequence.

Bonus question: What about the case where the coefficient ring is not a field?

Exercise 3.6. Compute the cohomology of ΩS^n , the loop space of the n -sphere. What is its cup product structure?

Exercise 3.7. If $E \rightarrow B$ is a fibration whose fiber is contractible, what can be said about the homology of E and B ? (For a spoiler of the answer, feed the fibration into the homotopy long exact sequence.) What about if B is contractible, or E is contractible?

Some more, from Nick:

Exercise 3.8. Prove the various universal coefficient theorems.

Exercise 3.9. Show that Ext and Tor are well-defined. That is, the two right derived functors which define $\text{Ext}_R^n(A, B)$ agree and the two left derived functors which define $\text{Tor}_n^R(A, B)$ agree.